Zero-Day Ransomware Behavioral Burst & Quarantine

Incident ID: INC-2026-0042 • **Incident Timestamp:** **10/08/2026, 15:11:25 IST** • **Classification:** INTERNAL SOC INCIDENT AUDIT

Target Endpoint:	`WS-FINANCE-01`	**Severity / Risk:**	**CRITICAL (94 / 100)**
IP Address:	`10.0.1.21`	**Containment Status:**	**CONTAINED (Quarantine Enforced)**
MAC Address:	`00:50:56:A1:B2:C1`	**Lead Analyst / System:**	SOC Lead Analyst & Sentinel Behavioral Kernel
Operating System:	Windows 11 Pro 23H2	**Response Latency:**	**< 11ms** (Minifilter Kill Hook)

1. EXECUTIVE SUMMARY

At 15:11:02 IST, Sentinel EDR intercepted a multi-stage zero-day ransomware attack attempting rapid in-place encryption across user documents and decoy honeytokens on WS-FINANCE-01. The rogue process (PID 4312) exhibited an anomalous rename velocity of 1,823 ops/min and surge in Shannon byte entropy to H=7.82. Within 11 milliseconds of touching Canary Trap TRAP-002, Sentinel's minifilter driver revoked the process file handles and placed the endpoint into an isolated network boundary. Zero corporate data was encrypted or exfiltrated.

2. AFFECTED ENDPOINT DETAILS

HOSTNAME	IP ADDRESS	OPERATING SYSTEM	COMPOSITE RISK
WS-FINANCE-01	**10.0.1.21**	**Windows 11 Pro 23H2**	**94 / 100 (CRITICAL)**

3. ATTACK CHAIN TIMELINE (CHRONOLOGICAL)

TIMESTAMP (IST)	INCIDENT PHASE	FORENSIC TELEMETRY & HEURISTIC OBSERVATION
`10/08/2026, 15:11:02 IST`	**Process Spawn**	PID 4312 (svchost_suspicious.exe) executed via PowerShell
`10/08/2026, 15:11:07 IST`	**File Burst**	1,823 file read/write operations/min detected in EDR_Test
`10/08/2026, 15:11:12 IST`	**Rename Velocity Spike**	Batch alteration (.locked extension appended)
`10/08/2026, 15:11:16 IST`	**Entropy Anomaly**	Shannon entropy H = 7.82 observed on honeypot file
`10/08/2026, 15:11:21 IST`	**Risk Escalation**	Dynamic composite score escalated to 78/100
`10/08/2026, 15:11:25 IST`	**Tripwire Hit & Containment**	Canary TRAP-002 accessed. Score 94/100. Host isolated.

4. DETECTION INDICATORS (IOCS)

INDICATOR TYPE	OBSERVED VALUE / THREAT INTELLIGENCE MATCH
SHA256	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`
C2 IP	`185.220.101.44 (Port 443 / 8443)`
MITRE ATT&CK;	`T1486 (Data Encrypted for Impact), T1490 (Shadow Copy Tampering)`
Canary Token	`CANARY_TOKEN_B6ED5D03 (Decoy Trap TRAP-002)`

5. CONTAINMENT ACTIONS TAKEN

[ENFORCED]	**Process handle revoked via Sentinel Minifilter**
[ENFORCED]	**Endpoint network isolated (NDIS quarantine filter applied)**
[ENFORCED]	**Outbound C2 sockets dropped**
[ENFORCED]	**Decoy honeypot integrity state updated to TAMPER_ATTEMPT_BLOCKED**

6. REMEDIATION RECOMMENDATIONS

- 1. Re-image WS-FINANCE-01 from verified pre-incident snapshot.
- 2. Enforce PowerShell Constrained Language Mode on executive workstations.
- 3. Block external egress to IP subnet 185.220.101.0/24 at firewall level.
- 4. Rotate active Canary Honeytoken secrets across Decoy vault.

Automated Defense Engine: Sentinel // EDR Prevention Kernel v2.4	**Incident Verification:** **CONTAINED & LOCKED**
Reporting Analyst: SOC Lead Analyst & Sentinel Behavioral Kernel	**Incident Timestamp: 10/08/2026, 15:11:25 IST**